



An Evaluation Framework for Temporal Information Extraction Systems

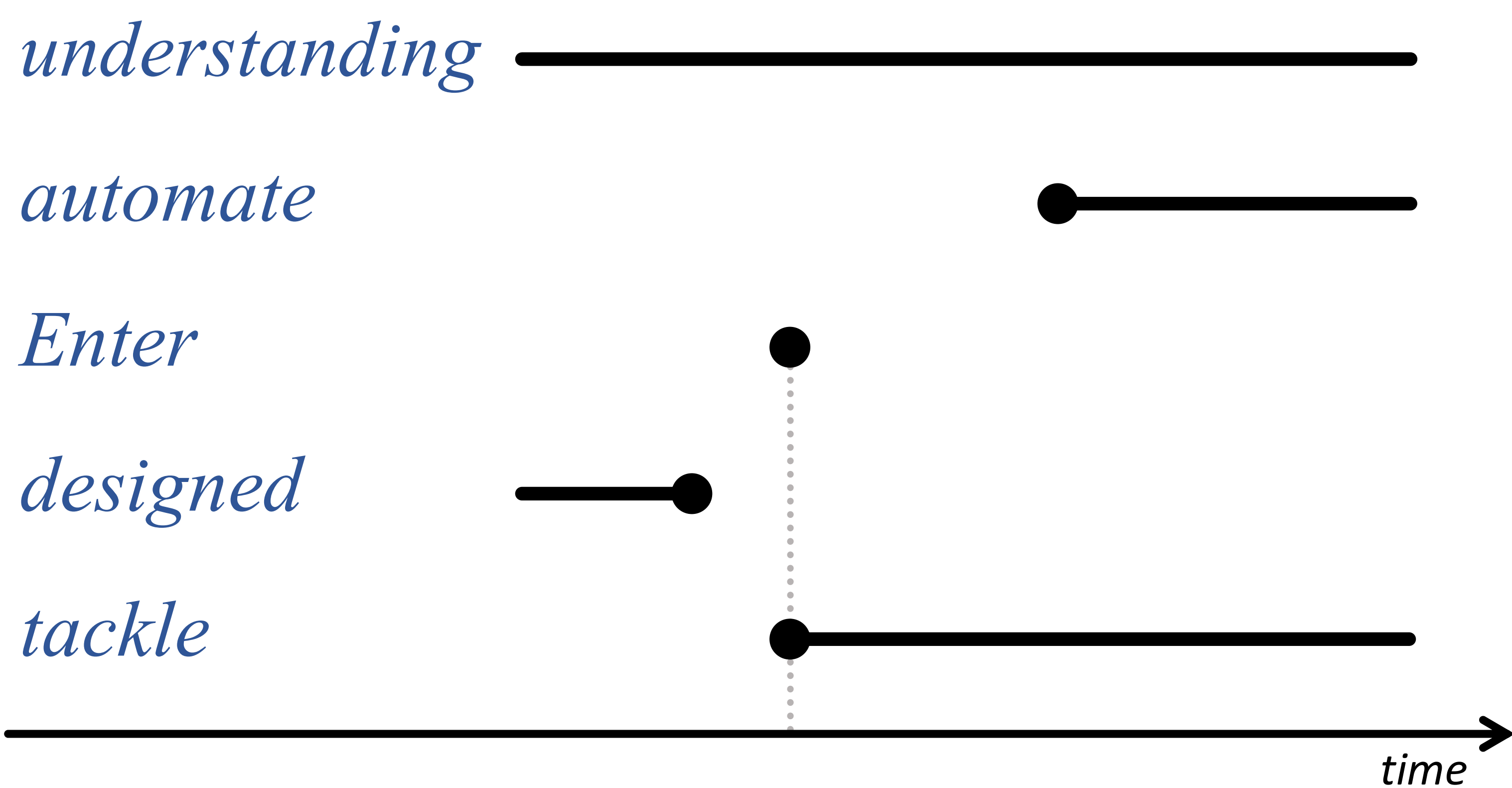
Hugo Sousa^{1,2}, Ricardo Campos^{1,3}, Alípio Jorge^{1,2}

¹ INESC TEC, Porto, Portugal ² University of Porto, Porto, Portugal ³ University Beira Interior, Covilhã, Portugal

In Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '23), July 23-27, 2023, Taipei, Taiwan.

“In a world where *understanding* the temporal order of events is essential, how can we *automate* such task? *Enter tieval*, a Python library *designed to tackle* this challenge.”

The task of **temporal information extraction** focus on the development of systems that can automatically understand the time, span, and temporal relations of these events. Having a system that can achieve human-level effectiveness on this task would find application in many natural language processing problems.



CHALLENGES

1. DATASET FORMAT
Datasets are released in different formats. Requiring a significant effort to compile and parse all of them.
2. DATASET PERSPECTIVE
Each of dataset has a different perspective. Ultimately preventing comparability among the different methods and corpora.
3. METRICS STANDARDIZATION
Lack of standardization in the metrics used to evaluate the systems. Mainly due to the lack of availability of domain-specific metrics.

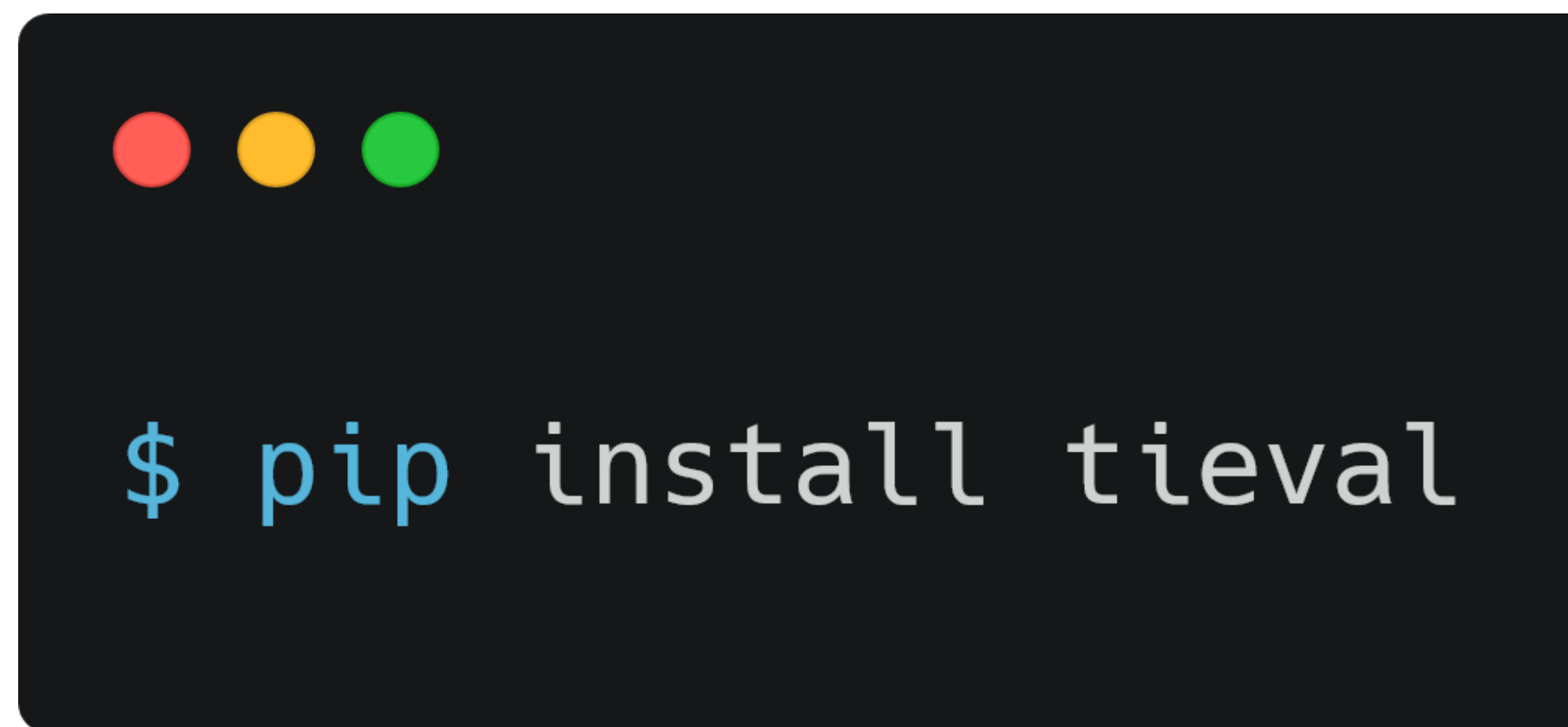
text2story

fct
Fundação
para a Ciência
e a Tecnologia

UNIVERSIDADE
BEIRA INTERIOR

U.PORTO

INESCTEC
TECHNOLOGY & SCIENCE



The vision for **tieval** was to build a framework that would support and facilitate the evaluation of temporal information extraction systems.

tieval contains **three modules** that represent the three cornerstones of any machine learning project: datasets, models, and evaluation. The **datasets** module is responsible for downloading and reading the corpora available for temporal information extraction, the **models** module is responsible for the construction of the models, and the **evaluate** module has methods to make a proper evaluation for each of the TIE tasks.

How can I load a dataset?

```
from tieval import datasets
te3 = datasets.read("TempEval_3")
```

Does it have models?

```
from tieval import models
model = model.TimexIdentificationBaseline()
model.fit(te3.train)
pred = model.predict(te3.test)
```

How to evaluate a model?

```
from tieval import evaluate
annt = {doc.name: doc.timexs for doc in te3.test}
result = evaluate.timex_identification(annt, pred)
```

What the results look like?

	f1	precision	recall
macro	0.778	0.817	0.742
micro	0.795	0.851	0.746

With **tieval**, we're providing the community with a standard way to benchmark and develop systems to unfold the temporal structure of text. Enabling the development of reproducible systems and pushing the boundaries of what's possible in Temporal Information Extraction.